

noteworthy that when working with codes, the terms are “decoded” (or “plaintext”, again) and “encoded” in that order.

Another term when it comes to coding in the context of cryptography is “codebooks”. In the simple case that we just presented, your life history serves as a codebook. To help you better understand, here’s another simple example:

Ethan and Fabio want to communicate with each other. But there’s a problem – Ethan knows only English while Fabio knows only French and can’t speak or understand English. Ethan too can’t understand French. In this case, Ethan would require a French->English dictionary so that when Fabio speaks in French, he could look up the words in the dictionary and find their meanings in English. Similarly, Fabio will need an English->French dictionary so that when Ethan speaks in English, he can lookup the words in the dictionary and understand their meanings in French. In this case, both the dictionaries serve as codebooks, using which one can find the “decoded” form of an “encoded” word. Technically, a codebook is a simple mapping of encoded-to-decoded words. If you were to transform all names in your phonebook in a way that no one could easily find out what name stood for which person, you might need a codebook other than your life history.

That was one form of encoding. The other form of encoding is called a “one-time” code. This form of code, as the name suggests, is used once only. One of the best examples of such code would be the military’s use of code-names for teams of soldiers. Let’s again consider an example: Assume that there’s a military general with three teams of soldiers. On the first day, he assigns names as: Team 1 => Alpha, Team 2 => Bravo and Team 3 => Charlie. Then he sends them out for patrol. The teams communicate using wireless radio devices whose signals can be intercepted. So when Team 1 has to talk to Team 2, a person from Team 1 says “Alpha to Bravo” to signal that the message is meant for Team 2 from Team 1. Though Team 3 also gets the message, they understand that the message is not for them. The next day, names are changed and Team 1 => Zebra, Team 2 => Delta and Team 3 => Tango. If Team 1 once again wants to convey a message to Team 2, they would say “Zebra to Delta”. If these messages were also being intercepted by enemy soldiers, they wouldn’t be able to determine the total number of teams (notice the first naming was done in alphabetical order, while the second was random). Such usage of codes is an example of a one-time code.

One-time codes are utilized only once (depending on what is defined as “once”) and are often used for messages which are to be “broadcasted” and